

Yajie Zhou

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EDUCATION

Nanchang University

Master of Medicine in Surgery

Nanchang, China

Sep 2024–Present

Jinggangshan University

Bachelor of Medicine in Clinical Medicine

Ji'an, China

Sep 2019–Jun 2024

RESEARCH INTERESTS

Cancer bioinformatics; computational oncology; tumor microenvironment remodeling; single-cell and spatial omics; tumor metabolism and mitophagy; lung adenocarcinoma and renal cancer; precision oncology.

PUBLICATIONS

Peer-reviewed Journal Articles

- Zhou, Y., Hu, Z., Xie, L., Zhang, W., & Rao, H. (2026). Comprehensive multi-omics profiling of tertiary lymphoid structures reveals immunogenetic landscapes and prognostic subtypes in lung adenocarcinoma. *Molecular Carcinogenesis*, 65(7), 888–903. <https://doi.org/10.1002/mc.70120>
- Zhou, Y., Hu, Z., Xie, L., Zhang, S., Huang, Q., Zhang, N., Liu, Y., Zhang, W., & Wang, K. (2026). Identification and external validation of a prognostic signature based on bone morphogenetic protein-related mRNAs for kidney renal clear cell carcinoma. *Discover Oncology*. <https://doi.org/10.1007/s12672-026-05238-z>
- Xie, L., Zhou, Y., Hu, Z., Zhang, W., & Zhang, X. (2025). Integrative multi-omics reveals energy metabolism-related prognostic signatures and immunogenetic landscapes in lung adenocarcinoma. *Frontiers in Immunology*, 16, 1679464. <https://doi.org/10.3389/fimmu.2025.1679464>

Manuscripts Under Review

- Zhou, Y., Hu, Z., Xie, L., Lai, Y., Kuang, J., Zhang, W., & Xie, L. Multi-omics characterization of ENO1-associated mitophagy–metabolism crosstalk and tumor microenvironment remodeling in lung adenocarcinoma. *Under review*.

RESEARCH EXPERIENCE

ENO1-associated Mitophagy–Metabolism Crosstalk and Tumor Microenvironment Remodeling in Lung Adenocarcinoma

Nanchang, China

Graduate Researcher, Nanchang University

2025–Present

- Developed an integrative multi-omics framework to investigate ENO1-associated metabolic epithelial states and tumor microenvironment remodeling in lung adenocarcinoma.
- Integrated bulk RNA-seq, survival modeling, WGCNA, pathway enrichment, single-cell RNA-seq, spatial transcriptomics, and CellChat-based cell–cell communication inference.
- Identified ENO1 as a candidate hub linking mitophagy-related transcriptional programs, metabolic reprogramming, malignant epithelial-state heterogeneity, and immune/stromal remodeling.
- Mapped ENO1 expression across epithelial subclusters and spatial tumor niches, generating experimentally testable hypotheses for ENO1 perturbation and protein-level validation.

TLS-related Immunogenetic Subtypes in Lung Adenocarcinoma

Nanchang, China

First Author, Published Project

2025–2026

- Constructed an integrated computational framework to characterize tertiary lymphoid structure-related molecular subtypes in lung adenocarcinoma.
- Combined Mendelian randomization, machine-learning survival modeling, immune microenvironment profiling, tumor mutational burden analysis, single-cell RNA-seq analysis, and RT-qPCR validation.
- Identified TLS-related prognostic subtypes with distinct immunogenetic landscapes, supporting patient stratification and tumor immune-state interpretation.

BMP-related mRNA Prognostic Modeling in Kidney Renal Clear Cell Carcinoma

Nanchang, China

First Author, Published Project

2025–2026

- Built and externally validated a bone morphogenetic protein-related mRNA prognostic signature for kidney renal clear cell carcinoma.
- Analyzed public cancer genomics cohorts to evaluate prognostic performance, tumor microenvironment features, tumor mutational burden, and drug-sensitivity associations.
- Linked BMP-related transcriptional patterns with immune contexture and clinical outcomes, supporting risk stratification in clear cell renal cell carcinoma.

Energy Metabolism-related Prognostic Signatures in Lung Adenocarcinoma

Nanchang, China

Co-author, Published Project

2024–2025

- Participated in an integrative multi-omics study of energy metabolism-related prognostic signatures and immunogenetic landscapes in lung adenocarcinoma.

- Contributed to transcriptomic analysis, immune microenvironment interpretation, prognostic model evaluation, figure organization, and manuscript preparation.

TECHNICAL SKILLS

- **Programming and Reproducible Research:** R, Python, Git/GitHub, Markdown, LaTeX.
- **Cancer Bioinformatics:** Bulk RNA-seq analysis, single-cell RNA-seq analysis, spatial transcriptomics, WGCNA, pathway enrichment analysis, survival analysis, prognostic modeling, tumor microenvironment analysis, tumor mutational burden analysis, drug-sensitivity analysis, multi-cohort validation.
- **Computational Methods:** Machine-learning-based survival modeling, epithelial subclustering, pseudotime analysis, immune/stromal compartment analysis, ligand–receptor interaction inference, and clinically interpretable multi-omics modeling.
- **Databases and Platforms:** TCGA, GEO, E-MTAB, public scRNA-seq datasets, public spatial transcriptomics datasets.
- **Experimental Validation Exposure:** RT-qPCR-based validation and experimentally testable hypothesis generation.

CLINICAL BACKGROUND

- Completed systematic medical training in physiology, pathology, pharmacology, internal medicine, surgery, and clinical reasoning.
- Built a clinically grounded perspective for interpreting cancer multi-omics results, prognostic models, and patient stratification frameworks.

LANGUAGES AND STANDARDIZED TESTS

- **Languages:** Mandarin Chinese, native; English, academic reading and scientific writing.
- **Standardized Tests:** TOEFL iBT: 5.5/6; GRE: 325.